The Explanation Is the Forward Pass: Interpretability by Construction with Certified Legible Bottlenecks

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Abstract

The dominant paradigm in interpretability is *archaeological*: train an opaque model, then dig for structure after the fact with probes, sparse autoencoders (SAEs), and attribution graphs. Each of these methods measures faithfulness *post hoc*; none *guarantees* it, and by 2026 the limits are well documented: SAEs do not recover canonical features and often fail to beat linear probes, attribution graphs leave unbounded computation in "error nodes", and chain-of-thought is an unfaithful, optimisation-fragile window. We argue that interpretability should instead be an *architectural invariant the model is required to satisfy*, with a guarantee emitted at inference time. We introduce the **Legible Bottleneck**, a single composable primitive that reads a layer's computation through a sparse, typed, persistently named concept code and reconstructs it; the unexplained remainder, the *leak*, is kept and measured rather than discarded. Stacking Legible Bottlenecks yields **CANDOR** (Concept-ANchored Disentangled Output Routing), a network co-trained so that the named concepts reconstruct the computation, are *causally sufficient* (interchange interventions as a training signal, not a post-hoc test), and keep anchored identities across training. On every forward pass, CANDOR emits a **Faithfulness Certificate** $\delta(x)$: the total-variation distance between the deployed model and its leak-ablated "legible" replacement. Because δ is a measurement rather than a learned monitor, it is sound by construction and re-checkable by any third party. On synthetic tasks with known ground-truth concepts and mechanism, CANDOR recovers the planted concepts (mean correlation 0.919), certifies its decisions with mean $\delta = 0.0021$ at no measurable interpretability tax, and passes held-out causal tests (81.7% directional accuracy); a backdoor that bypasses the concept channel is exposed by a certificate spike (detection AUC 0.901). The primitive composes with attention. On the pretrained GPT-2 the certificate machinery transfers, but neither a *frozen* retrofit nor single-layer fine-tuning favours CANDOR over a reconstruction-only SAE (a negative that is stable across seeds and loss weightings), and an $\approx 8\times$ reconstruction improvement under fine-tuning fails to tighten the certificate: reconstruction and faithfulness dissociate. Training a small language model from scratch *with* its bottlenecks reverses the picture, as the thesis predicts: on real text the full objective certifies materially tighter than a matched reconstruction-only control ($\delta = 0.4667$ vs. 0.6454) at no measured capability tax, though the absolute certificate remains loose; whether by-construction training certifies *tightly* at realistic scale is the central open question. We are explicit about scope: these are small-scale demonstrations, and completeness is unattainable under computation in superposition; the contribution is to make the bypass *measured, bounded, and certified.* All code and every measured number are released.

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1 Introduction

For a decade interpretability has been practised as *archaeology*: train a model whose computation is opaque, and then, with linear probes, sparse autoencoders, activation patching, attribution graphs, and chain-of-thought inspection, excavate. The excavations are remarkable: induction heads [Olsson et al., 2022], a circuit for indirect object identification [Wang et al., 2023], millions of monosemantic features [Templeton et al., 2024], and per-prompt attribution graphs that traced planning and multi-step reasoning in a production model [Lindsey et al., 2025]. But archaeology has a structural problem that scaling does not fix: it produces *descriptions* of a model, validated *after the fact*, never a *guarantee* that the description is what the model computes.

By 2026 the limits are no longer speculative. Sparse autoencoders (SAEs), the field's workhorse, were shown not to find canonical units of analysis [Leask et al., 2025], to suffer feature absorption and hedging [Chanin et al., 2024, 2025], and, decisively, to *not* reliably beat linear probes or even random baselines on the supervised tasks one would want them for [Kantamneni et al., 2025]. Attribution graphs, the state of the art in tracing computation, route an admitted and unbounded fraction of the work through "error nodes" that the explanation simply does not cover [Ameisen et al., 2025]. Chain-of-thought, the most accessible window of all, is unfaithful [Turpin et al., 2023, Chen et al., 2025] and, worse, becomes *actively misleading* under optimisation pressure [Baker et al., 2025, Korbak et al., 2025]. The common root is that every one of these methods measures faithfulness post hoc; none makes the model *be* faithful.

Thesis. An advanced agent whose decisions must be trusted cannot be audited by digging. We propose the opposite stance: make legibility a property the architecture is *required* to have, and emit the audit *at inference time*. Concretely, force the output-relevant computation through a sparse, typed, persistently named concept channel, co-train that channel to be *causally* faithful, and measure, on every forward pass, how much computation escaped it. The escaped fraction is not hidden; it is the certificate. This reframes the central question from "can one find structure in the model?" to "how much of the model's decision is *not* accounted for by named concepts, and can the model prove a bound on that number right now?"

The primitive. The central object is the **Legible Bottleneck** (§3): at a checkpoint with activation h, encode a sparse non-negative code $c = \mathrm{TopK}(\mathrm{ReLU}(W_{\mathrm{enc}}(h - b_{\mathrm{dec}})))$, decode $\hat{h} = W_{\mathrm{dec}}c + b_{\mathrm{dec}}$, and route the network with the *leak* $r = h - \hat{h}$ kept and measured rather than thrown away. It is one operation, it composes, and, like self-attention [Vaswani et al., 2017] for sequence modelling, the architecture is just the primitive, stacked. We call the stack **CANDOR** (Concept-ANchored Disentangled Output Routing, §4).

The guarantee. The deployed model M has a twin, the *legible replacement* M_{leg}, obtained by ablating every leak; M_{leg}'s output is a pure function of the named concept codes. The **Faithfulness Certificate** is $\delta(x) = \mathrm{TV}(M(x), M_{\mathrm{leg}}(x))$, the total-variation distance between the deployed model and its named-concept explanation (§6). Crucially δ is *measured*, not predicted by a learned head, so it cannot be Goodharted into silence the way a learned monitor can [Baker et al., 2025]: it is a sound, one-sided, per-input, re-checkable upper bound on the influence of all non-legible ("dark") computation.

Contributions.

1. **The Legible Bottleneck** (§3) and the **CANDOR** architecture (§4): a by-construction realisation of "the explanation is the forward pass" in which output-relevant computation is routed through a sparse, typed, anchored concept channel and the unexplained remainder is a first-class, measured quantity.
2. **A training objective** (§5) that conjoins completeness, *causal* faithfulness (interchange and causal-scrubbing interventions as a training signal), and concept anchoring; we show empirically that the conjunction is necessary: removing any term makes the certificate vacuous or the channel leak.
3. **The Faithfulness Certificate** (§6): a per-forward-pass, sound-by-construction, re-checkable bound on dark computation, with the precise predicate it does and does not guarantee stated honestly (Propositions 1–3).
4. **A ground-truth evaluation, and an honest real-LLM probe** (§7): on synthetic tasks with known concepts and known mechanism, where soundness can actually be checked, CANDOR recovers the concepts, certifies tightly at near-zero tax, passes a directional causal test, detects a channel-bypassing backdoor via a certificate spike, and composes with attention on a sequence task. On the pretrained GPT-2 we show the certificate machinery transfers to a real LLM *and*, honestly, that neither a frozen retrofit nor single-layer fine-tuning favours CANDOR over an SAE (stably across seeds and weightings), while an $\approx 8\times$ reconstruction improvement fails to tighten the certificate, dissociating reconstruction from faithfulness and delimiting the contribution to the by-construction regime. In that regime, at the smallest honest scale, the prediction holds: a language model trained from scratch with its bottlenecks certifies materially tighter than its matched reconstruction-only control ($\delta = 0.4667$ vs. 0.6454) at no measured tax, the thesis's first real-text support, with an absolute certificate that remains loose. Every number is measured and released.

We do not claim to have solved interpretable AI. We claim a concrete, checkable mechanism that converts interpretability from an after-the-fact description into an architectural invariant with a runtime guarantee, together with an honest account of exactly how far that guarantee currently reaches.

2 Background: six stages toward legibility

The modern interpretability literature reads as a sequence of attempts to make neural computation legible, each born from the failure of the last; CANDOR sits at the end of the arc and is best understood as a conjunction of its final three stages.

Stage 0: the substrate (superposition). Networks pack many more sparse, near-linear features than they have neurons [Elhage et al., 2022, 2021, Olah et al., 2020], which both motivates a sparse legible basis and bounds it: if *computation* (not just storage) occurs in superposition [Hänni et al., 2024], no fixed-width legible channel can be complete. We concede this up front.

Stage 1: recover features post hoc (SAEs). Dictionary learning disentangles superposition into nameable features [Bricken et al., 2023, Cunningham et al., 2023, Templeton et al., 2024, Gao et al., 2024], with architectural refinements [Rajamanoharan et al., 2024, Bussmann et al., 2024, 2025] and open suites [Lieberum et al., 2024]. The 2025 crisis [Leask et al., 2025, Kantamneni et al., 2025, Chanin et al., 2024] diagnoses the flaw CANDOR targets: *no SAE objective is causally faithful by construction*, and reconstruction error is large and behaviourally load-bearing.

Stage 2: trace computation (transcoders, attribution graphs). Transcoders approximate a layer's input-output map [Dunefsky et al., 2024]; cross-layer transcoders plus replacement models yield per-prompt attribution graphs [Ameisen et al., 2025, Lindsey et al., 2025, Marks et al., 2025]. This is the closest prior art to "the explanation is the forward pass." Its limits (unbounded error nodes, per-prompt cost, post-hoc construction) are exactly what a runtime certificate must fix.

Stage 3: make faithfulness causal. The field converges on intervention as the only credible test: activation/path patching [Goldowsky-Dill et al., 2023, Zhang and Nanda, 2024], automated discovery [Conmy et al., 2023, Syed et al., 2023], causal scrubbing [Chan et al., 2022], and the unifying theory of causal abstraction [Geiger et al., 2021, 2025]. Decisively, faithfulness can be a *training* objective: interchange intervention training (IIT) [Geiger et al., 2022] and distributed alignment search [Geiger et al., 2024, Wu et al., 2023]. This is CANDOR's core borrowed engine.

Stage 4: build legibility in. Force computation through a privileged basis: concept bottleneck models [Koh et al., 2020, Espinosa Zarlenga et al., 2022], self-explaining networks [Alvarez-Melis and Jaakkola, 2018], prototypes [Chen et al., 2019], B-cos alignment [Böhle et al., 2022], codebooks [van den Oord et al., 2017], the information bottleneck [Tishby and Zaslavsky, 2015], and the call to use inherently interpretable models for high-stakes decisions [Rudin, 2019]. Their recurring failure is *concept leakage*: the bottleneck silently re-becomes a black box [Mahinpei et al., 2021, Havasi et al., 2022], and identifiability is unattainable without inductive bias [Locatello et al., 2019].

Stage 5: certify. Neural-network verification is mature for *robustness* [Katz et al., 2017, Zhang et al., 2018, Wang et al., 2021, Xu et al., 2020] but robustness is not interpretability; formal-XAI gives per-input logical guarantees [Ignatiev et al., 2019]; interpretable-by-construction-and-verified models exist only at small scale [Przybysz et al., 2023, Granmo, 2018, Lin et al., 2020]. No method emits a *sound, cheap, per-forward-pass* bound on bypassed computation.

CANDOR is the conjunction of Stages 3, 4 and 5 performed *jointly* and emitted *at runtime*. §8 states precisely what is borrowed from each and what is new.

3 The Legible Bottleneck

Let $h \in \mathbb{R}^d$ be the activation at a chosen checkpoint (e.g. an MLP hidden state). A Legible Bottleneck is parameterised by an overcomplete dictionary of $m \gg d$ atoms.

Definition 1 (Legible Bottleneck). *With encoder $W_{\text{enc}} \in \mathbb{R}^{m\times d}$, decoder $W_{\text{dec}} \in \mathbb{R}^{d\times m}$ (unit-norm columns), and biases $b_{\text{enc}}, b_{\text{dec}}$, define*

$$c = \operatorname{TopK}_k(\operatorname{ReLU}(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})) \in \mathbb{R}^m_{\geq 0}, \qquad \hat{h} = W_{\text{dec}}\, c + b_{\text{dec}}, \qquad r = h - \hat{h}.$$

c is the concept code *(at most k active named concepts), $\hat{h}$ the* legible reconstruction*, and r the* leak*.*

The bottleneck does not replace h; it *routes* the downstream network in one of three modes, and this routing, not the dictionary, is what separates the construction from a post-hoc sparse code:

- **full** (deployed): the network receives $h = \hat{h} + r$ unchanged. The bottleneck is an auxiliary read plus a regulariser, so capability is *never* destroyed by construction; the interpretability tax can only enter through training pressure, not through a lossy forward pass.

- **legible** (replacement): the network receives $\hat{h}$; the leak is ablated. The resulting model M_{leg} has an output that is a pure function of the named concept codes.
- **leak-swap** (causal probe): the network receives $\hat{h} + r_\pi$, the leak replaced by another input's leak under a permutation π. If the named concepts are causally sufficient, the output is unchanged; the training objective drives this.

The TopK non-linearity (cf. TopK SAEs [Gao et al., 2024]) gives an exact $L_0 = k$ budget (a hard, legible cap on how many concepts may participate in a single decision), while unit-norm decoder atoms supply geometric identifiability. Each atom carries a persistent integer identity (a *name*) and an optional *type* tag; §5 keeps those identities stable across training.

4 The CANDOR architecture

A CANDOR model places Legible Bottlenecks at the checkpoints that carry output-relevant computation. In the reference implementation we study (i) an MLP whose concept layer sits on a frozen, identity-initialised embedding of the input (the site at which generative factors are linearly present, so the sparse code can disentangle them), followed by a task head, and (ii) a Transformer whose every MLP hidden state is a bottleneck site, demonstrating composition with attention. The deployed model M and its legible twin M_{leg} share all weights and differ only in routing mode, so M_{leg} is not a separate distilled model fitted after the fact (as a transcoder replacement model is [Ameisen et al., 2025]) but the *same network* read through its concept channel. The explanation of a decision is then literally the list of active named concepts and their causal contributions, available at inference with no search.

5 The training objective

CANDOR is trained with one loss whose terms are mutually enabling. Writing ℓ_{task} for cross-entropy, sites for the per-layer $(h, \hat{h}, r, c)$, and π for a random batch permutation,

$$\mathcal{L} = \underbrace{\ell_{\text{task}}(M)}_{\text{accurate}} + \alpha \underbrace{\ell_{\text{task}}(M_{\text{leg}})}_{\text{legible model accurate}} + \beta \underbrace{\mathrm{KL}(M \,\|\, M_{\text{leg}})}_{\text{faithfulness}}$$
$$+ \lambda \underbrace{\textstyle\sum_\ell \frac{\|r^{(\ell)}\|^2}{\|h^{(\ell)}\|^2}}_{\text{leak energy}} + \mu \underbrace{\mathrm{TV}(M,\, M_{\text{leak-swap}(\pi)})}_{\text{causal sufficiency}} + \gamma \underbrace{\textstyle\sum_\ell \|W^{(\ell)}_{\text{dec}} - \bar{W}^{(\ell)}\|^2}_{\text{anchoring}}.$$

(The implementation additionally applies a mild L_1 shrinkage to the concept codes; Appendix A lists every weight.) The *faithfulness* term trains the leak-ablated replacement to reproduce the deployed model's behaviour, which drives the certificate δ toward zero. The *leak-energy* term pulls the activation onto the sparse-decodable manifold so the named code actually explains h. The *causal-sufficiency* term is the load-bearing, borrowed-from-IIT ingredient [Geiger et al., 2022, Chan et al., 2022]: it penalises the output's sensitivity to swapping the (non-legible) leak across inputs, i.e. it trains the concepts to be a causal, not merely correlational, summary. The *anchoring* term penalises drift of each decoder atom from its persistent EMA identity $\bar{W}$, the inductive bias that identifiability theory proves necessary [Locatello et al., 2019] and that SAEs notably lack [Leask et al., 2025]. Objective ablations (§7) show each term is necessary: drop faithfulness or leak and δ becomes large (the channel leaks); drop causal and the leak regains causal power; drop anchoring and concept identities stop being stable across runs.

Listing 1: The Legible Bottleneck and the CANDOR step (reference implementation).

```
def legible_bottleneck(h, W_enc, b_enc, W_dec, b_dec, k):
    c = topk(relu(W_enc @ (h - b_dec) + b_enc), k) # sparse named concept code
```

```
    h_hat = W_dec @ c + b_dec # legible reconstruction
    return c, h_hat, h - h_hat # code, recon, leak

# one training step (3 forwards: deployed, legible, leak-swapped)
L_full = task(M(x, mode="full"), y)
L_leg = task(M(x, mode="legible"), y)
faith = KL(M(x,"full").detach(), M(x,"legible")) # replacement reproduces model
leak = mean(||r||^2 / ||h||^2 for sites) # shrink dark computation
causal = TV(M(x,"full").detach(), M(x,"full", leak_perm=randperm)) # leak is inert
loss = L_full + a*L_leg + b*faith + lam*leak + mu*causal + gam*anchor
```

6 The Faithfulness Certificate

Definition 2 (Faithfulness Certificate). *For input x, let $p_M(\cdot \mid x)$ and $p_{\text{leg}}(\cdot \mid x)$ be the output distributions of the deployed model and its legible replacement. The certificate is*

$$\delta(x) \;=\; \mathrm{TV}\big(p_M(\cdot \mid x), p_{\text{leg}}(\cdot \mid x)\big) \;=\; \tfrac{1}{2}\sum_y \big|p_M(y \mid x) - p_{\text{leg}}(y \mid x)\big| \;\in [0,1].$$

Proposition 1 (Soundness, by construction). *$p_{\text{leg}}(\cdot \mid x)$ depends on x only through the named concept codes $\{c^{(\ell)}(x)\}$. Hence for every event A over outputs, $\big|p_M(A \mid x) - p_{\text{leg}}(A \mid x)\big| \le \delta(x)$: the named-concept explanation predicts the deployed model's entire output distribution to within $\delta(x)$. Because $\delta(x)$ is* computed *by one extra (legible) forward pass rather than predicted by a learned head, it is exact and one-sided: it can never* under-*report behavioural divergence between the model and its explanation.*

Proof sketch. Total variation equals $\sup_A |p_M(A) - p_{\text{leg}}(A)|$, giving the bound for all A. M_{leg} ignores every leak by definition, so its output is a deterministic function of $\{c^{(\ell)}\}$. $\delta(x)$ is obtained by evaluating both forward passes and summing absolute differences; no parameter is fit to it, so it is a direct measurement. □

Proposition 2 (Causal sufficiency ⇔ leak-inertness). *Write the deployed output as $f(c, r)$ for concept code c and leak r. If for all inputs the output is invariant under replacing r by any other in-distribution leak r' (leak-swap invariance), then f does not depend on r on the data support, i.e. the output is determined by the named concepts alone. The causal term in $\mathcal{L}$ minimises exactly the leak-swap output deviation; the residual deviation and δ jointly quantify the failure of sufficiency.*

Proof sketch. For any two in-distribution leaks r, r' compatible with a code c, invariance gives $f(c, r) = f(c, r')$, so $f(c, \cdot)$ is constant on the support and the output is a function of c alone. The causal term is the expected output deviation under a random leak permutation, a Monte Carlo estimate of the violation of this invariance. □

Proposition 3 (Checkability). *$\delta(x)$ and the explanation (active named concepts and, via concept ablation, their causal effects) are recomputable by any third party from the released weights with one extra forward pass per input, without access to training data or process. The certificate is therefore independently auditable.*

Proof sketch. Every quantity in the definitions above is a function of the released weights and the input alone: p_M and p_{leg} require one standard and one legible-mode forward pass, the active concepts are read off $c(x)$, and per-concept causal effects need one additional legible pass per ablated concept. □

Remark 1 (No completeness claim). *CANDOR does* not *claim $\delta \equiv 0$ is attainable at scale. Computation in superposition [Hänni et al., 2024] implies some output-relevant work has no*

Quantity	**CANDOR**	**Opaque + post-hoc SAE**
Task accuracy (deployed)	99.9%	99.9%
Task accuracy (legible / replacement)	99.9%	(n/a)
Interpretability tax	≈ 0 pp	(n/a)
Faithfulness certificate δ (mean)	0.0021	none emitted
model/explanation agreement	99.9%	(n/a)
Concept recovery (vs. ground truth)	0.919	0.966
Causal necessity gap	77.3 pp	(n/a)
Directional causal accuracy	81.7%	(n/a)
Non-legible remainder, causal effect	0.0054	0.0030
Per-input certificate	emitted	none

Table 1: Planted-concepts results. CANDOR attains concept recovery comparable to the standard train-then-excavate pipeline (a post-hoc SAE on the opaque model) at no measurable accuracy tax. The difference is *guarantees*: only CANDOR emits a per-input faithfulness certificate (δ), trains its concepts to be causally sufficient (the held-out necessity and directional tests), and, shown in the ablations, anchors their identity across runs. On this easily-reconstructed task the SAE's uncaptured residual happens to be nearly inert too (0.0030 vs. CANDOR's 0.0054), so incidental inertness is *not* the differentiator here; the certificate and the trained causal properties are. The regime where the certificate becomes decisive is computation that *bypasses* the channel (§7.2), which the post-hoc pipeline has no certificate to catch.

faithful sparse-linear description, so a non-trivial bypass can persist. The value of the certificate is soundness, not tightness*: a large but honest $\delta(x)$ tells an auditor exactly when* not *to trust the named-concept explanation, which is precisely the signal post-hoc methods cannot provide. This also reframes the certificate as a constructive answer to eliciting latent knowledge [Christiano et al., 2021] and a quantitative mechanistic-anomaly detector [Roger et al., 2023]: dark computation is surfaced as a number, not hidden in an error node.*

7 Experiments

Following the discipline that interpretability claims are only meaningful where they can be *checked*, our primary evaluation is on synthetic data with known ground-truth concepts and a known mechanism. We then show composition with attention on a sequence task, and probe a real pretrained LLM (GPT-2). All numbers are measured by `experiments/run_all.py` and emitted to `paper/_numbers.tex`; nothing is hand-entered. Appendix A lists every architecture, hyperparameter, and seed.

7.1 Planted concepts: a checkable ground truth

We draw $K = 12$ independent sparse binary concepts, mix them linearly (in superposition) into a 48-dimensional observation, and set the label to the number of active concepts among a fixed relevant subset of size 5 (a 6-way task). Because the true concepts *and* the true mechanism are known, recovery, causal faithfulness, and certificate soundness are all directly checkable. We train a CANDOR-MLP (dictionary $m = 48$, $k = 8$, 2200 steps) and, for comparison, an opaque MLP of matched shape with a post-hoc TopK SAE fitted to its embedding, the standard "train-then-excavate" pipeline.

Table 1 summarises the headline result. CANDOR's legible replacement solves the task at 99.9%, as accurate as the opaque baseline (99.9%), i.e. *no measurable interpretability tax*, while certifying every decision with a mean $\delta = 0.0021$ (95th percentile 0.0024) and agreeing with its own named-concept explanation 99.9% of the time. It recovers the 12 planted concepts (mean correlation 0.919, minimum 0.763); the post-hoc SAE recovers them slightly better (0.966).

Recovery alone, however, does not differentiate the pipelines: both find the concepts. What CANDOR adds is everything that turns a *description* into a *guarantee*: a per-input certificate (the SAE pipeline emits none), causal sufficiency trained into the channel, anchored identity, and the ability to detect channel-bypassing computation (§7.2). On this clean task the SAE's uncaptured residual is itself nearly inert (0.0030, comparable to CANDOR's 0.0054), so incidental inertness is not what separates them here; the certificate and the trained causal properties are. Labelling activations after the fact can recover concepts, but it cannot certify that they are what the model used.

Causal faithfulness. On a *held-out* intervention the channel was not trained on (ablating the named concept matched to a relevant generative factor and asking whether the decision moves by exactly the amount the true mechanism dictates), CANDOR is 81.7% directionally correct on relevant concepts and leaves the decision unchanged 99.6% of the time when an irrelevant concept is ablated. The necessity gap (the decision-flip rate when the atoms matched to the relevant concepts are ablated, versus an equal number of irrelevant atoms) is 77.3 percentage points (77.3% vs. 0.0%).

Objective ablations (Fig. 3). Each term removes a *distinct* guarantee, which is exactly why the conjunction, not any single part, is the contribution. Dropping the *leak-energy* term collapses concept recovery from 0.919 to 0.496: the code stops explaining the activation, even though δ stays small. Dropping the *causal* term keeps δ small but restores the leak's causal power (leak-swap effect 0.0110, roughly double CANDOR's 0.0054). This is behavioural mimicry without causal sufficiency, the exact failure a non-causal, reconstruction-only method cannot even detect. Dropping *anchoring* leaves within-run metrics intact but, across independently trained runs, concept-identity stability falls from 0.722 to 0.000, reproducing the SAE non-canonicality finding [Leask et al., 2025] and showing anchoring fixes it. The *faithfulness* term keeps the certificate tight (removing it raises δ modestly, to 0.0030). No single objective yields completeness, causal sufficiency, stable identity, and a tight certificate together; their conjunction does.

7.2 Dark-computation detection: a backdoor that cannot hide (Fig. 2)

We plant a rare trigger (4.5% of inputs), a direction *disjoint* from the named concepts, that flips the label. To use it, the model must route the computation through the leak, so the certificate should react. It does: δ on triggered inputs (mean 0.0040) is $\approx 6\times$ that on clean inputs (0.0006), and δ alone detects the backdoor with AUC 0.901 (the model itself reaches 100.0% accuracy, so the backdoor is fully operational). This is the central anti-Goodhart evidence: because δ is a measurement of behavioural divergence and not a learned monitor, computation that bypasses the named channel *cannot* be hidden while remaining functional: it either becomes a named concept (inspectable) or shows up as dark computation (a δ spike). This is the property that optimisation-fragile chain-of-thought monitors lack [Baker et al., 2025].

7.3 Composition with attention

The Legible Bottleneck is architecture-agnostic. To show it composes with self-attention, we train a one-layer Transformer with a Legible Bottleneck on its MLP hidden state on a task that *requires* attention: predict the maximum token of a length-8 sequence (10 classes, 10200 train / 1800 test), which can only be solved by aggregating information across positions. CANDOR's deployed model reaches 99.9% and its leak-ablated legible replacement 99.9% (an interpretability tax of 0.00 percentage points), while certifying with mean $\delta = 0.0000$ and 100.0% model/explanation agreement. The Legible Bottleneck composes with self-attention without collapsing capability; a full study of larger algorithmic circuits (e.g. modular addition and induction) is left to future work.

7.4 A real pretrained LLM: certificate transfer and the limits of retrofitting

Does any of this survive contact with a real language model? We retrofit a Legible Bottleneck as a transcoder on one MLP layer (9 / 12) of the pretrained **GPT-2 small** (124M parameters), on real text, and measure the certificate by *actually splicing the channel into GPT-2* and running the real model (its own causal attention, the rest of its layers) on held-out text. We train two channels at matched sparsity ($m = 2048$, $k = 32$): a reconstruction-only **SAE** and a **CANDOR** channel (reconstruction + behavioural faithfulness + causal sufficiency).

There are two findings, and we report the second prominently because it is the more instructive. *First, the certificate machinery transfers to a real LLM:* splicing the channel and running GPT-2 yields a measurable, sound, per-input δ (total variation at the next token) of ≈ 0.1136–0.1299 at this sparsity, so the runtime certificate of §6 is not confined to toy models. *Second, in the frozen-retrofit regime the reconstruction-only SAE is the better channel:* it is modestly more behaviourally faithful ($\delta = 0.1136$ vs. CANDOR's 0.1299), more causally inert (leak-swap 0.1319 vs. 0.1597), and reconstructs better (unexplained variance 0.375 vs. 0.486), and this ordering is stable across loss weightings, so it is structural, not a tuning artefact.

The reason, we hypothesised, is the thesis of this paper. On a *frozen* model the only lever that shrinks the leak is better reconstruction, precisely the SAE's objective. CANDOR's distinctive objectives (causal sufficiency, anchoring) assume the model can *reshape its own computation* to route through the legible channel; a frozen retrofit cannot, so they only spend capacity. The favourable tax/faithfulness frontier of Experiments 7.1–7.3 comes from training the model *with* the bottleneck, *by construction*, not from the dictionary alone. This delimits the contribution cleanly: CANDOR is not a drop-in replacement for a post-hoc SAE. It also yields a testable prediction: give the model room to reshape its computation and the behavioural and causal objectives should begin to pay. We test the smallest version of that prediction next.

Layer-wise fine-tuning: a dissociation, and the prediction does not yet hold. We unfreeze the spliced layer's MLP (4.7M of 124M parameters) and fine-tune it jointly with the channel on the same text for 500 steps, under the full language-modelling loss (which anchors capability), in two conditions matched in every respect (unfrozen parameters, steps, batch, seed, learning rates; Appendix A) except the channel objective: reconstruction only (**FT-SAE**) versus reconstruction + behavioural faithfulness + causal sufficiency (**FT-CANDOR**). Two findings. *First, reconstruction and faithfulness dissociate.* Fine-tuning drives the control's unexplained variance from 0.375 to 0.046, an $\approx 8\times$ improvement, yet its certificate does not tighten ($\delta = 0.1282$, versus 0.1136 for the frozen retrofit; both fine-tuned models also adapt to the corpus, from 5.443 to ≈ 4.194 validation loss, so frozen-to-fine-tuned numbers compare channels on different underlying models, while the within-regime comparisons are exactly matched). Near-perfect reconstruction buys no behavioural faithfulness, which is precisely the gap between reconstruction-style objectives and guarantees that motivates this paper. *Second, the prediction does not hold at this scale.* At matched language-modelling loss (4.225 vs. 4.194), FT-CANDOR certifies *less* tightly than the control ($\delta = 0.1560$ vs. 0.1282) and its leak is less inert (0.1723 vs. 0.1485). Reshaping a single MLP under a joint objective, at this budget, is not the regime in which the behavioural and causal signals win; we report this negative result at the same standard as the positives (every number measured and released). A sweep over 3 seeds and a tenfold range of objective weightings ($\beta = \mu \in \{0.3, 1.0, 3.0\}$; Appendix A) confirms the negative is structural rather than an artefact of the seed or the weighting: FT-SAE certifies at $\delta = 0.1283 \pm 0.0012$ (mean $\pm$ standard deviation over seeds) while FT-CANDOR reaches 0.1654 ± 0.0090, 0.1609 ± 0.0131 and 0.1429 ± 0.0075 at the three weightings. Raising the weights tightens FT-CANDOR's certificate somewhat but pays a language-modelling cost (4.313 vs.

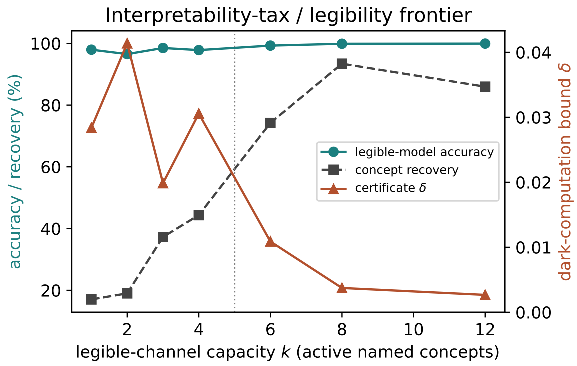

Figure 1: The interpretability-tax / legibility frontier. As the legible channel's capacity k (active named concepts) grows past the task's true concept count (dotted line), legible-model accuracy and concept recovery saturate while the certificate δ collapses: once the channel can carry the task's concepts, legibility is nearly free.

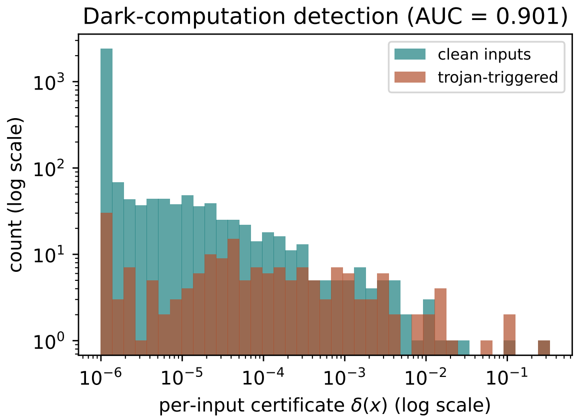

Figure 2: Dark-computation detection. The per-input certificate $\delta(x)$ separates clean inputs from backdoor-triggered inputs whose computation bypasses the named channel (AUC 0.901). A bypass cannot be both functional and hidden.

4.189 validation loss at the heaviest weighting), and no setting closes the gap to the control. The by-construction gains of Experiments 7.1–7.3 have therefore *not* been reproduced on a pretrained model, by retrofitting or by local fine-tuning; the decisive experiment is *full* by-construction training, co-training the network with its bottlenecks from the start, which we run next (§7.5).

7.5 By-construction training on real text

The retrofit and fine-tuning results delimit where CANDOR's objectives do *not* pay; neither tests the thesis, which is a claim about *training*. Here we run the decisive experiment at the smallest honest scale: a language model trained from scratch *with* its bottlenecks. Three conditions share one architecture (a 4-layer decoder-only Transformer, $d_{\text{model}} = 256$, 4 heads, MLP width 1024), one corpus (TinyShakespeare under GPT-2 tokenisation: 2376 training and 264 validation sequences of length 128), one optimiser, batch schedule and step budget, and 2 seeds (Appendix A). **OPAQUE** has no bottlenecks: the capability reference and the substrate for the post-hoc pipeline (TopK SAEs at matched $m = 1024$, $k = 32$, fitted to its MLP activations afterwards and spliced back in, at the final site as in §7.4 and at every site for an ablation-matched comparison). **RECON-ONLY** places a Legible Bottleneck on every MLP hidden state and trains with the LM loss plus leak energy only: the control that separates "a reconstructing channel is present" from "the CANDOR objectives act". **CANDOR** is the full conjoined objective.

The by-construction prediction holds here, on every faithfulness metric, in both seeds. At matched architecture and training, CANDOR certifies materially tighter than the control ($\delta = 0.4667 \pm 0.0049$ vs. 0.6454 ± 0.0062, mean $\pm$ standard deviation over seeds), its leak is more causally inert (leak-swap 0.5787 vs. 0.7479), its legible replacement agrees with the deployed model on 48.0% of held-out next-token decisions against 31.6%, and its legible model is the more capable one (validation loss 7.998 vs. 8.251). The control is not a strawman: it reconstructs *better* than CANDOR (unexplained variance 0.082 vs. 0.189), yet its certificate ($\delta = 0.6454$) is barely tighter than post-hoc SAEs spliced into the frozen opaque model at every site ($\delta = 0.6574$). The dissociation of §7.4 therefore runs in both directions: reconstruction gains bought no tightening

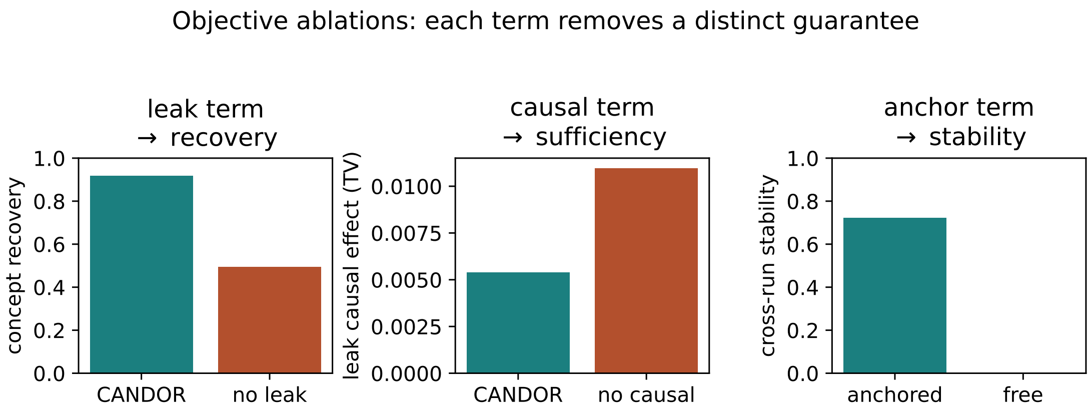

Figure 3: Objective ablations: each term is load-bearing for a *distinct* guarantee. **Left:** the leak-energy term governs concept recovery (dropping it halves recovery). **Centre:** the causal term governs causal sufficiency (dropping it roughly doubles the leak's causal effect, mimicry without sufficiency). **Right:** the anchoring term governs cross-run concept-identity stability (dropping it sends it to zero). No single objective delivers all three; the conjunction does.

there, and here CANDOR trades reconstruction away for behavioural faithfulness. The ordering is unchanged under the single-site protocol of §7.4, ablating only the final site (CANDOR 0.2515, control 0.3907, post-hoc SAE 0.3668).

There is no measured tax at this scale; the sign reverses. CANDOR's full model generalises better than the opaque reference (validation loss 8.528 vs. 9.600; RECON-ONLY sits between at 9.168). Every condition overfits this small corpus (training loss near zero), so we read the reversed sign as a regularisation effect of the bottleneck objectives in the overfitting regime, measured honestly, not as evidence that legibility is free at scale.

The absolute certificate stays loose. CANDOR's mean δ of 0.4667 is more than two orders of magnitude above the synthetic-task regime (§7.1): on real text at this scale, roughly half of the output distribution's mass is still shaped by computation the named concepts do not carry, and the certificate says so on every forward pass. By-construction training buys a *materially tighter* certificate than any retrofit, control, or post-hoc splice we measured; it does not yet buy a *tight* one. This gives the thesis its first real-text support and sharpens the open question: whether the gap widens or closes at realistic scale and data.

8 Related work

We state precedents prominently: *every component of CANDOR has antecedents*, and the novelty is their conjunction operating jointly and at runtime.

Superposition and the legibility substrate. [Elhage et al., 2022, 2021, Olah et al., 2020, Park et al., 2024, Hänni et al., 2024]. CANDOR inherits both the motivation for a sparse legible basis and the completeness limit it concedes (Rem. 1).

Sparse dictionaries and their crisis. [Bricken et al., 2023, Cunningham et al., 2023, Templeton et al., 2024, Gao et al., 2024, Rajamanoharan et al., 2024, Bussmann et al., 2025, Lieberum et al., 2024, Chanin et al., 2024, Leask et al., 2025, Kantamneni et al., 2025]. CANDOR borrows the sparse typed channel and the reconstruction (completeness) objective; it departs by making the channel causal and certified rather than post-hoc, and by anchoring identity.

Transcoders, circuits, attribution. [Dunefsky et al., 2024, Marks et al., 2025, Ameisen et al., 2025, Lindsey et al., 2025, Wang et al., 2023, Conmy et al., 2023, Syed et al., 2023, Miller et al., 2024]. CANDOR borrows "the explanation is the forward pass" and error-node accounting; it departs by routing *into* the trained forward pass and bounding the error-node mass at runtime.

Causal faithfulness as an objective. [Geiger et al., 2021, 2022, 2024, Wu et al., 2023, Chan et al., 2022, Geiger et al., 2025, Makelov et al., 2024]. This is CANDOR's core borrowed engine. CANDOR generalises IIT from aligning a *given* causal variable to co-training a typed, persistent, *complete* bottleneck plus a runtime certificate; it does not invent training-time causal faithfulness.

By-construction interpretability and its leakage. [Koh et al., 2020, Alvarez-Melis and Jaakkola, 2018, Espinosa Zarlenga et al., 2022, Chen et al., 2019, Böhle et al., 2022, van den Oord et al., 2017, Hewitt et al., 2023, Rudin, 2019, Mahinpei et al., 2021, Havasi et al., 2022, Locatello et al., 2019]. CANDOR is a concept bottleneck hardened with the guarantees this line proved necessary; *leakage is dark computation under another name*, and the certificate is what bounds it.

Certification, oversight, evaluation, governance. [Katz et al., 2017, Zhang et al., 2018, Xu et al., 2020, Przybysz et al., 2023, Ignatiev et al., 2019, Christiano et al., 2021, Roger et al., 2023, Lanham et al., 2023, Chen et al., 2025, Baker et al., 2025, Korbak et al., 2025, Jacovi and Goldberg, 2020, Karvonen et al., 2025, Amodei, 2025, Clymer et al., 2024, European Union, 2024]. CANDOR borrows the sound-over-approximation form and the safety-case framing, and supplies the runtime certificate that monitoring and CoT approaches cannot. The runtime-certificate stance also continues a line of *certifying* runtime guardrails for deployed models, but moves the certificate from the model's *outputs* to the model's own *computation*.

9 Limitations and honest scope

- **Scale, and the retrofit/by-construction gap.** Our tightly certified evidence is on synthetic ground-truth tasks and a one-layer Transformer, *not* a frontier LLM; we chose the regime where soundness can be *checked*. On the real pretrained GPT-2 (§7.4) the certificate *machinery* transfers, but neither a *frozen* retrofit nor single-layer joint fine-tuning favours CANDOR over a reconstruction-only SAE (stable across three seeds and a tenfold weighting range), and under fine-tuning an $\approx 8\times$ reconstruction improvement fails to tighten the certificate. The hypothesis that CANDOR's gains require the model to reshape its computation through the channel failed in every retrofit regime and held in the smallest by-construction one (§7.5): trained from scratch with its bottlenecks, a four-layer LM certifies materially tighter than its matched reconstruction-only control at no measured tax, but its absolute certificate on held-out text remains loose ($\delta = 0.4667$), the comparison spans two seeds and one small corpus, and the reversed tax is an overfitting-regime effect. The single most important next experiment is therefore by-construction training at realistic scale and data, followed by a frontier-scale evaluation (SAEBench, RAVEL, MIB, and a real backdoor-detection battery [Karvonen et al., 2025, Huang et al., 2024, Mueller et al., 2025]); these are future work, not present claims. Whether the certificate tightens rather than loosens as scale and capability grow is the central open empirical question.
- **No completeness.** By Rem. 1 we do not and cannot claim the certificate reaches zero on arbitrary computation; we claim it is sound, measured, and trainable, and that a large δ is an honest abstention signal.

- **"Certificate" $\neq$ formal proof.** $\delta(x)$ is a sound *per-input measurement*, not a verified bound over all inputs. It is one-sided (cannot under-report divergence for the inputs evaluated) but says nothing, on its own, about inputs not run. A periodic exact certificate on a tractable channel fragment [Przybysz et al., 2023] and a worst-case bound-propagation audit [Xu et al., 2020] are the natural complements.
- **Anchoring tension.** Anchoring injects inductive bias to stabilise identity; pushed too hard it makes concepts the designer's labels rather than the model's. We report where CANDOR sits via the cross-run stability number rather than assuming the best case.
- **Concept placement.** On the planted task we place the concept layer where the generative factors are linearly present; identifying such sites automatically in a deep agent is unsolved, and computation that is irreducibly distributed across many sites will show up as leak, not as a clean concept.

10 Broader impact

This work is about transparency and oversight of capable AI. A runtime, sound, checkable bound on a model's non-legible computation is exactly the kind of artefact a safety case for an autonomous agent needs [Clymer et al., 2024, Amodei, 2025] and that transparency regulation increasingly asks for [European Union, 2024]. The same mechanism is dual-use only in the weak sense that any interpretability tool is: it makes models easier to understand, audit, and correct. The honest risk we flag is *false assurance*: treating a low δ on a test distribution as a guarantee off-distribution. We therefore frame δ as a measurement with stated assumptions, demonstrate that it *rises* on bypassing computation rather than staying silent, and release everything needed to recompute it.

11 Conclusion

Interpretability has been practised as archaeology: build an opaque artefact, then dig. We have argued that the way toward interpretable AI systems is to stop digging, to make legibility an architectural invariant and emit the audit at inference time. The Legible Bottleneck is the single primitive that makes this possible: route a layer's computation through a sparse, typed, anchored concept code, keep and *measure* the unexplained leak, and train the concepts to be causally sufficient. The resulting CANDOR architecture certifies each of its decisions with a sound, re-checkable bound on its own dark computation, recovers ground-truth concepts at near-zero tax, and surfaces a channel-bypassing backdoor that no post-hoc method bounds. It is not a solution to interpretable AI, and it does not claim completeness. It is a concrete, checkable mechanism for turning interpretability from a description one hopes is true into an invariant the model proves, one forward pass at a time.

Reproducibility statement

Every number in this paper is machine-generated: `experiments/run_all.py` (and the LLM scripts `experiments/exp_gpt2.py`, `exp_gpt2_ft.py`, `exp_gpt2_ft_sweep.py`, `exp_lm_scratch.py` for §7.4–7.5) writes raw results to `results/*.json`, `scripts/paper_numbers.py` converts them into the macros used in the text (`paper/_numbers.tex`), and `scripts/make_figures.py` renders the figures. No number is hand-entered, and `make all` reproduces the paper end-to-end from a fresh checkout. All random seeds are fixed in the experiment scripts (Appendix A); the synthetic-task experiments run on CPU, the LLM experiments on a single commodity GPU, and all of them use only publicly available weights and text. Code, the reference implementation (`candorkit`), unit tests, and all measured results are available under an MIT license at https://github.com/AnwarDebes/candor.

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A Experimental details

This appendix records every architecture, hyperparameter, and seed, as fixed in the released code. Optimisation is plain Adam throughout; no learning-rate schedules, weight decay, data augmentation, or early stopping are used anywhere. The anchoring reference $\bar{W}$ is an exponential moving average of the decoder with momentum 0.99, updated once per step.

A.1 Planted concepts (§7.1)

Data. 20,000 samples; $K = 12$ independent Bernoulli concepts with density 0.3, mixed linearly into 48 dimensions with additive Gaussian noise ($\sigma = 0.1$); the label counts the active concepts among a fixed relevant subset of size 5 (6 classes); generation seed 0; 80/20 train/validation split (seed 0).

Model. CANDOR-MLP: the concept layer sits on a frozen, identity-initialised embedding of the input; dictionary $m = 48$, $k = 8$; task head with two hidden layers of width 128.

Training. 2200 steps, Adam, learning rate 2×10^{-3}, batch size 512, seed 0. Loss weights: $\alpha = 0.5$ (legible task), $\beta = 1.0$ (faithfulness), $\lambda = 1.0$ (leak energy), $\mu = 0.5$ (causal sufficiency), $\gamma = 10^{-2}$ (anchoring), plus an L_1 code penalty of 2×10^{-3}.

Baseline. An opaque MLP of matched shape (width 128, two hidden layers) trained for the same 2200 steps (learning rate 2×10^{-3}, batch 512), followed by a post-hoc TopK SAE ($m = 48$, $k = 8$) fitted to its embedding for 2,000 steps (Adam, learning rate 10^{-3}, batch 1,024, MSE reconstruction, decoder columns renormalised each step).

Causal tests. Atoms are matched to generative concepts by Hungarian assignment on absolute Pearson correlation. The *necessity* test ablates (zeroes) the atoms matched to the relevant concepts and compares the legible decision-flip rate against ablating an equal number of irrelevant atoms. The *directional* test ablates one relevant concept's atom at a time, on the inputs where that concept is active, and checks that the legible decision moves by exactly the amount the ground-truth mechanism dictates (one class down, for the sum task), and symmetrically that ablating an irrelevant concept's atom leaves the decision unchanged. Both tests run on up to 1,500 held-out validation inputs; neither intervention appears in the training objective.

Ablations. Each ablation retrains from the same initialisation (seed 0) with one weight zeroed: no-faithfulness ($\beta = 0$), no-leak ($\lambda = 0$), no-causal ($\mu = 0$), no-anchoring ($\gamma = 0$); remaining weights $\beta = 1.0$, $\lambda = 1.0$, $\mu = 0.5$, $\gamma = 10^{-2}$ as applicable, L_1 penalty 10^{-3}.

Frontier sweep (Fig. 1). The tax/legibility frontier retrains the same architecture at channel capacities $k \in \{1, 2, 3, 4, 6, 8, 12\}$ for 1,600 steps each (learning rate 2×10^{-3}, batch 512, seed 0) with the default loss weights ($\alpha = 0.5$, $\beta = 1.0$, $\lambda = 0.3$, $\mu = 0.3$, $\gamma = 10^{-2}$, L_1 10^{-3}), on the same data and split as the main run.

Stability protocol. Three training runs (seeds 11, 22, 33) per condition; the *anchored* condition uses anchoring weight 10^{-2} and a shared dictionary initialisation (the anchor's reference point), the *free* condition uses no anchoring and per-seed initialisation; remaining weights at their defaults ($\alpha = 0.5$, $\beta = 1.0$, $\lambda = 0.3$, $\mu = 0.3$, L_1 10^{-3}). Reported stability is the mean pairwise concept-identity agreement across runs.

Backdoor (§7.2). 20,000 samples with $K = 12$ concepts of which 3 are relevant, density 0.3, noise 0.1; the base task is the parity of the relevant concepts (binary). The trigger is a fixed unit direction in 4 extra input dimensions (total input dimension 52), disjoint from the concept mixing directions by construction, added with magnitude 3.0 against background noise $\sigma = 0.1$; it is planted in a nominal 5% of inputs (measured 4.5% of the validation split) and flips the label. The model (same architecture) is trained with the default loss weights for 2200 steps; detection AUC uses the per-input $\delta(x)$ as the score, with no access to trigger labels.

A.2 Sequence task (§7.3)

Data. 12,000 sequences of length 8 over a vocabulary of 10 tokens (generation seed 0); the label is the maximum token (10 classes); 85/15 split (10200 train / 1800 test).

Model. One-layer Transformer: $d_{\text{model}} = 96$, 4 attention heads, MLP width 256, with a Legible Bottleneck ($m = 128$, $k = 8$) on the MLP hidden state.

Training. 1,200 steps, Adam, learning rate 10^{-3}, batch size 256, seed 0; loss weights $\alpha = 0.5$, $\beta = 1.0$, $\lambda = 0.5$, $\mu = 0.3$, $\gamma = 10^{-2}$, L_1 penalty 10^{-3}; trained on CPU.

A.3 GPT-2 retrofit (§7.4)

Setup. GPT-2 small (124M parameters, frozen throughout); a transcoder-style Legible Bottleneck reads the MLP input and predicts the MLP output at layer 9 of 12, spliced into the live model via a forward hook. Corpus: the public TinyShakespeare text, tokenised with the GPT-2 tokeniser into 1,400 sequences of length 24; 85/15 split (1,190 train / 210 validation sequences).

Channels. Both channels use $m = 2048$, $k = 32$, unit-norm decoder columns, and identical training budgets: 250 steps, Adam, learning rate 2×10^{-3}, batch size 12, seed 0, on CPU. The *SAE* channel minimises normalised MSE reconstruction of the MLP output only. The *CANDOR* channel adds the behavioural faithfulness term (KL between the spliced legible model's next-token distribution and the cached full-model distribution) and the causal-sufficiency term (total variation against the leak-swapped splice), each with weight 1.0.

Metrics. δ is the mean total-variation distance between the full and legible-spliced next-token distributions on held-out text; the leak-swap effect is the same quantity against the leak-swapped splice; unexplained variance is residual energy over output energy; the loss-recovered fraction compares the legible splice's language modelling loss against the full model and a zero-ablated MLP.

A.4 GPT-2 layer-wise fine-tuning (§7.4)

Setup. Same model, layer, corpus, tokenisation, and 85/15 split as the frozen retrofit. The MLP of the spliced layer (4.7M parameters) is unfrozen; all other model parameters stay frozen. The model is kept in evaluation mode throughout, so dropout is off (evaluation mode does not block gradients).

Conditions. Both conditions train the channel ($m = 2048$, $k = 32$) jointly with the unfrozen MLP for 500 steps (Adam; channel learning rate 2×10^{-3}, MLP learning rate 10^{-4}; batch 12; seed 0) under the full language-modelling cross-entropy over all positions, which anchors capability, plus the normalised reconstruction loss with a *live* (non-detached) target, so in both conditions the MLP may reshape its output toward reconstructability. The FT-CANDOR condition adds the behavioural faithfulness term (KL between the spliced legible model's next-token distribution and the current full model's) and the causal-sufficiency term (total variation against the leak-swapped splice), each with weight 1.0; FT-SAE adds nothing. Matched step counts mean FT-CANDOR spends roughly three forward passes per step to FT-SAE's one, as in the frozen comparison. Evaluation is identical to the frozen experiment, and the original model's validation loss (5.443) is reported as the capability reference. The headline run uses seed 0 and weight 1.0 on both behavioural terms.

Robustness sweep. `experiments/exp_gpt2_ft_sweep.py` repeats the identical protocol over seeds $\{0, 1, 2\}$ and weightings $\beta = \mu \in \{0.3, 1.0, 3.0\}$ (applied jointly to the faithfulness and causal terms); FT-SAE does not depend on the weighting and is trained once per seed. Sweep numbers in §7.4 are means and standard deviations over the three seeds; raw per-run values are in `results/gpt2_ft_sweep.json`.

A.5 By-construction LM training (§7.5)

Data. The public TinyShakespeare text, tokenised with the GPT-2 tokeniser into non-overlapping sequences of length 128, shuffled with a fixed corpus seed (0) and split 90/10 by sequence (2376 train / 264 validation sequences); the identical data and split are used by every condition and seed.

Architecture. All conditions share a 4-layer pre-norm decoder-only Transformer: $d_{\text{model}} = 256$, 4 attention heads, MLP width 1024, learned positional embeddings, untied unembedding, next-token logits at every position. OPAQUE has no bottlenecks. RECON-ONLY and CANDOR place a Legible Bottleneck ($m = 1024$, $k = 32$) on every MLP hidden state (4 sites); in mode `full` the bottlenecks do not alter the computation, so all three conditions compute the same function class.

Training. 4000 steps, Adam, learning rate 3×10^{-4}, batch size 32, plain fp32 (no mixed precision), no learning-rate schedule, weight decay, or dropout; seeds $\{0, 1\}$, with the batch schedule drawn from the same seed-matched generator in every condition, so the three conditions see the same data in the same order. Bottleneck conditions renormalise decoder columns and update the anchoring EMA (momentum 0.99) each step.

Objectives. OPAQUE: next-token cross-entropy only. RECON-ONLY: cross-entropy plus the leak-energy term with weight $\lambda = 0.3$; no behavioural, causal, anchoring, or sparsity terms, so it isolates "a reconstructing channel is present" from "the CANDOR objectives act". CANDOR: the full conjoined objective at the library defaults, identical across seeds: $\alpha = 0.5$ (legible task), $\beta = 1.0$ (faithfulness KL between the full and legible next-token distributions, averaged over every position), $\lambda = 0.3$ (leak energy), $\mu = 0.3$ (causal sufficiency, total variation against the leak-swapped forward, averaged over every position), $\gamma = 10^{-2}$ (anchoring), L_1 penalty 10^{-3}.

Post-hoc pipeline. After OPAQUE training, a TopK SAE with matched $m = 1024$, $k = 32$ is fitted to each block's MLP hidden states over the training split (4000 steps, Adam, learning rate 10^{-3}, batch 1,024 rows, MSE reconstruction, decoder columns renormalised each step), then spliced into the frozen opaque model by loading its weights into a bottleneck twin (verified exactly identical in mode `full`) and routing through the fitted dictionaries. It is evaluated both at the final site only, matching the single-site retrofit protocol of §7.4, and at every site, the ablation-matched comparison to the by-construction conditions.

Metrics. As everywhere else: δ and the leak-swap effect are total variation between next-token distributions, here averaged over all positions of the held-out split; agreement is top-1; unexplained variance is residual energy over hidden-state energy, per site. The tax is the difference in full-model validation loss against OPAQUE.

A.6 Compute

All experiments run on a single commodity machine (the LLM experiments use one GPU); no data-centre resources are used. Each `results/*.json` file records the device and wall-clock time of the run that produced it.